

Machine Learning Practical Assignment

Q1: Use the Breast Cancer Wisconsin dataset to build an SVM classifier that predicts whether a tumor is benign or malignant and perform the following:

- Preprocess the data (handle missing values, normalization, encoding if needed).
- Split the dataset into training and test sets.
- Train an SVM model using different kernels (linear, RBF, polynomial).
- Evaluate the model performance using accuracy, precision, recall, and F1-score.
- Perform hyperparameter tuning using GridSearchCV.

Q2: Use the Iris dataset to build a decision tree classifier to predict the species of the flower and perform the following:

- Split the data into training and test sets.
- Train a Decision Tree classifier and visualize the tree.
- Perform hyperparameter tuning using GridSearchCV.

Q3: Use the Titanic dataset to build a random forest model that predicts passenger survival and perform the following:

- Perform exploratory data analysis and feature engineering.
- Handle categorical variables and missing data.
- Train a Random Forest classifier and evaluate it using confusion matrix
- Compare performance with a Decision Tree classifier.

Q4: Use the Wine Quality dataset to build a KNN model to classify wine quality and perform the following:

- Normalize the dataset.
- Choose the appropriate value of K using cross-validation.
- Train and test the model.
- Plot accuracy vs different values of K.

- Evaluate the model using appropriate metrics (confusion matrix, classification report).

Q5: Apply K-Means Clustering on the Iris dataset (without using species labels) to group the flowers based on their features and perform the following:

- Load the dataset and remove target labels.
- Standardize the features using StandardScaler.
- Use the Elbow method and Silhouette Score to determine the optimal number of clusters.
- Train a K-Means model and visualize the clusters in 2D

Q6: Use Hierarchical Agglomerative Clustering to analyze a dataset of customer spending habits (Mall Customers dataset:

<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>) and perform the following:

- Load the dataset and preprocess it.
- Compute and plot the dendrogram to determine the number of clusters.
- Apply AgglomerativeClustering using different linkage methods: ward, average, complete.

Q7: Implement K-Means and Fuzzy C-Means clustering algorithms from scratch to the below dataset

x	y
1.0	2.0
1.5	1.8
2.0	2.5
8.0	8.0
8.5	8.5
9.0	7.5